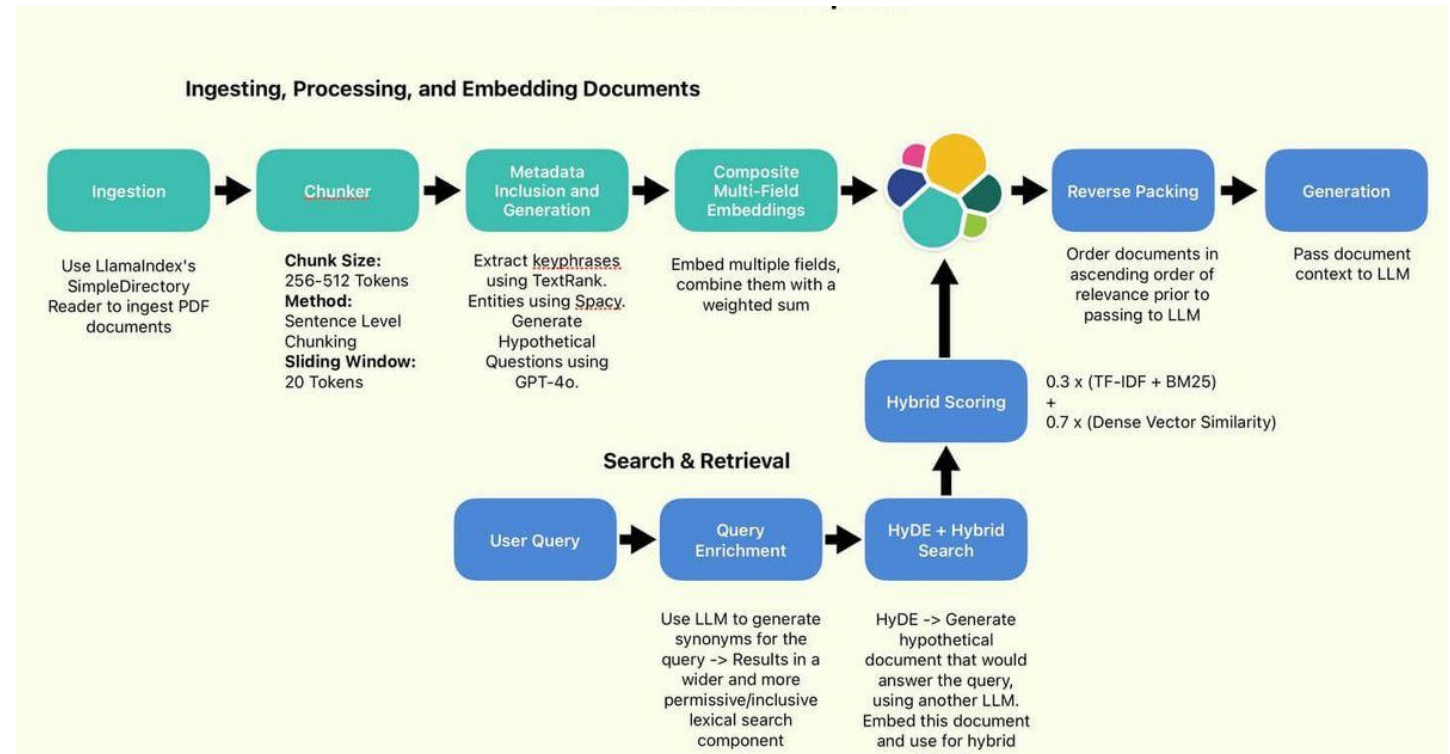
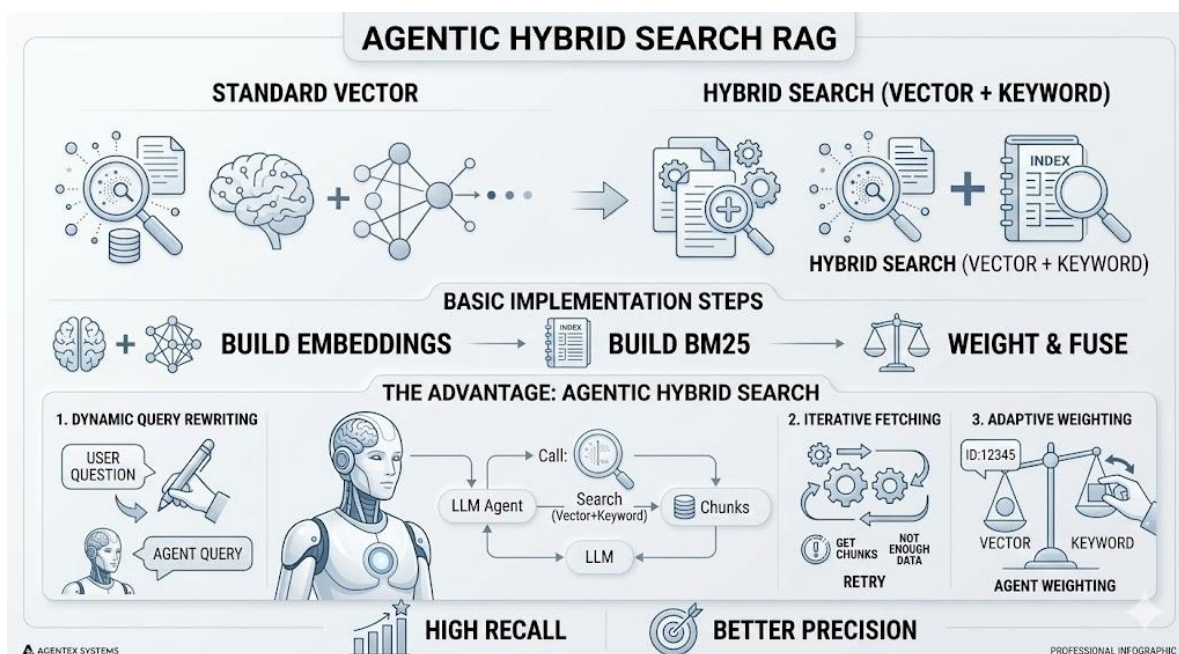


Module 4 – Advanced RAG & Vector Engineering



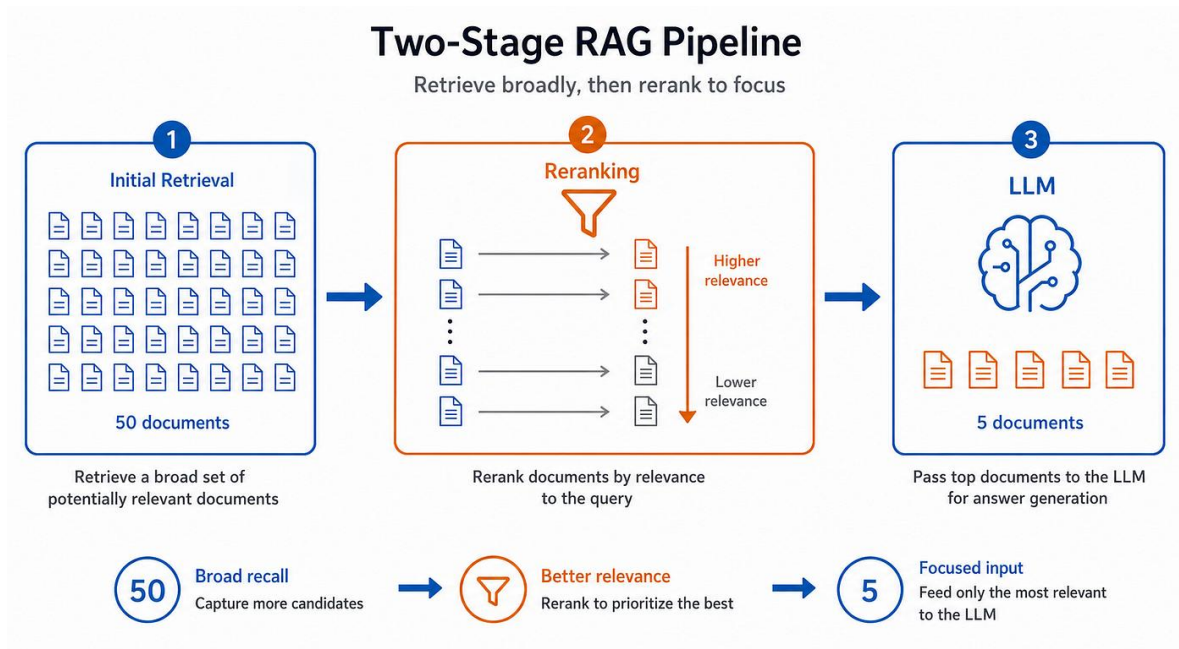
- **Goal:** Eliminating hallucinations and ensuring document traceability.
- **The Problem:** Standard RAG often misses context or retrieves "noisy" data that confuses the model.
- **The Solution:** Advanced Vector Engineering (Hybrid Search + Reranking) and Agentic reasoning.

Hybrid Search & Agentic RAG



- **Hybrid Search:** Combining **Keyword (BM25)** for exact matches (like serial numbers or medical terms) and **Vector Search** for semantic meaning.
- **Agentic RAG:** Instead of a single "retrieve and answer" step, the agent can look at the results and say: *"This isn't enough information. I will try a different search query."*

Reranking & Evaluation



- **Reranking:** Using a **Cross-Encoder** to analyze the top 50 results and pick the best 3. It's slower but much more accurate than a standard vector search.
- **RAGAS Framework:** Using AI to grade the AI. We measure:
 - **Faithfulness:** Is the answer derived *only* from the context?
 - **Relevancy:** Does it actually answer the user's question?
- **Task 4 Deliverable:** A "Memory Vault" that provides answers with specific citations (e.g., "Source: manual.pdf, Page 12").